

■ Email Domain Inspector

Setup & Usage Guide

By Jason Carter · github.com/jasonscottcarter

Overview

Email Domain Inspector is a Node.js web application that inspects email infrastructure for any domain or mail server IP. It performs DNS lookups for MX, SPF, DKIM, DMARC, BIMI, MTA-STS, and TLS-RPT records, checks the server IP against 10 major DNS blacklists, and presents everything in a clean dark-themed dashboard — all using free public DNS. No API keys required.

Prerequisites

Install the following tools before cloning the repository. Verify each one in a terminal after installation.

Node.js (v18 or later)	Download from nodejs.org — choose the LTS version. Verify: <code>node --version</code>
Git	Download from git-scm.com — use all defaults during install. Verify: <code>git --version</code>
VS Code (optional)	Recommended editor. Download from code.visualstudio.com .

No API keys are needed. All lookups are DNS-based or use the Node.js built-in `https` module.

Installation

1 — Clone the repository

```
git clone https://github.com/jasonscottcarter/email-inspector.git
cd email-inspector
```

2 — Install dependencies

```
npm install
```

This installs Express — the only external package. You should see found 0 vulnerabilities when complete.

3 — Start the server

```
node server.js
```

You should see: Email Inspector running at `http://localhost:3000`

4 — Open the dashboard

Open your browser and go to `http://localhost:3000`. To stop the server, press `Ctrl+C` in the terminal.

Usage

Enter any domain or mail server IP into the search box and click Inspect. The tool auto-detects whether you entered a domain or IP and queries the appropriate sources.

Domain	Enter a bare domain — e.g. gmail.com or proofpoint.com Returns: MX, SPF, DKIM, DMARC, BIMI, MTA-STS, TLS-RPT, RBL check
Mail Server IP	Enter a raw IPv4 address — e.g. 74.125.141.27 Returns: Reverse DNS lookup + RBL blacklist check

What Each Check Returns

MX Records	Mail exchanger hostnames and their priority values. Lower priority = preferred server.
SPF	Sender Policy Framework. Shows the full record and policy analysis (hard fail, softfail, neutral, or dangerous pass-all).
DKIM	DomainKeys Identified Mail. Auto-tries 16 common selectors covering Microsoft 365, Google Workspace, Mimecast, Proofpoint, and generic configurations. Reports which selectors were found and their public key records.
DMARC	Domain-based Message Authentication. Shows policy (none / quarantine / reject), subdomain policy, enforcement percentage, DKIM and SPF alignment mode, and aggregate/forensic report destinations.
BIMI	Brand Indicators for Message Identification. Shows logo URL, VMC authority URL, and renders the brand logo inline if present.
MTA-STS	Mail Transfer Agent Strict Transport Security. Fetches both the DNS record and the hosted policy file. Shows mode (enforce / testing / none), max age, and the list of allowed MX hosts.
TLS-RPT	TLS Reporting. Shows where TLS failure reports are sent (rua destination).
RBL Check	Checks the resolved mail server IP against 10 major DNS blacklists including Spamhaus, Barracuda, SpamCop, SORBS, UCEProtect, and Mailspike.

■ **DKIM selector discovery is inherently limited — there is no standard DNS mechanism to enumerate all selectors for a domain. If a domain uses a custom selector not in the built-in list, it will not be found. This is expected behavior, not a bug.**

Project File Structure

```
email-inspector/
```

```
■■■ node_modules/ # Installed automatically by npm install

■■■ public/

■ ■■■ index.html # Frontend dashboard

■■■ .gitignore

■■■ package.json

■■■ package-lock.json

■■■ server.js # Express backend + all DNS logic
```

Troubleshooting

node: command not found	Node.js is not installed or not in your PATH. Reinstall from nodejs.org and open a new terminal window before retrying.
Cannot find module 'express'	Dependencies not installed. Run <code>npm install</code> from inside the email-inspector folder.
Port 3000 already in use	Another process is using port 3000. Open <code>server.js</code> , change 3000 to 3001 at the bottom of the file, save, restart the server, and access the app at <code>http://localhost:3001</code> .
All RBL checks show errors	Your network or DNS resolver may be blocking outbound DNS queries to RBL zones. Try a different network or DNS server.
MTA-STS policy not fetched	The domain has a DNS record but the policy file at <code>https://mta-sts./well-known/mta-sts.txt</code> is unreachable. This is a configuration issue on the domain owner's side.

Email Domain Inspector is open source. Contributions welcome at github.com/jasonscottcarter/email-inspector